

Manoj Mareedu

Open to relocation • +91 7995453399 • manoj.mareedu.pro@gmail.com • linkedin.com/in/manojmareedu • <https://github.com/ManojMareedu>

AI /ML Engineer

An IEEE-published AI/ML Engineer and Data Scientist who ships complete production systems, not just notebooks. I've built the same disciplined MLOps stack (ZenML, MLflow, DVC, FastAPI, Docker, Streamlit) across three different problems: fraud detection, healthcare cost prediction, and retail demand forecasting, and each time found and fixed a real methodology bug instead of reporting an inflated number. As a freelancer, I've taken a pharmaceutical distributor in Tanzania from a manual Excel workflow to a fully offline desktop app their recovery team uses daily, and built FinSight AI, a multi-agent GenAI platform that researches public companies end to end. Master's in Business Analytics, UT Dallas. Open to global AI/ML and Data Science roles.

WORK EXPERIENCE

Freelance Consulting

08/2024 - Present

AI/ML Engineer • Freelancer

- Deployed FinSight AI, a multi-agent due-diligence platform that researches any US public company end to end, by building a LangGraph agent graph with conditional routing and RAG over SEC 10-K filings via ChromaDB.
- Gave full visibility into every LLM call, by integrating Langfuse tracing to capture token usage, latency, and risk scores per inference
- Replaced a Tanzanian pharmaceutical distributor's manual weekly Excel review with a real-time credit-risk dashboard, by building a Python ETL pipeline (pandas, watchdog) that reprocesses their accounting export within seconds of every save
- Gave the client's recovery team a ranked, always-current worklist instead of a static spreadsheet, by designing a weighted risk score across invoice age and amount and a four-tier priority system for hundreds of overdue invoices
- Delivered the tool as a fully offline native app for Windows and Mac, by packaging a Streamlit dashboard inside a pywebview shell with PyInstaller, so the client needs no server, account, or internet connection

The University of Texas at Dallas

05/2023 - 05/2024

Data Scientist (Student Assistant) • Part-time

Dallas, TX, USA

- Kept department leadership working from reliable numbers, by pulling daily Salesforce exports into a shared SharePoint folder that fed the team's live Power BI dashboard
- Caught data quality issues before they reached a report, by reviewing that data through exploratory analysis and flagging irregular program records to the right stakeholder
- Ran logistics end to end for department events, from procurement and vendor purchases through on-site coordination
- Split time between front-desk and analytical duties in a hybrid student-assistant role, supporting both day-to-day office operations and BI/marketing reporting

Cognizant Technology Solutions

08/2021 - 07/2022

Programmer Analyst • Full-time

India

- Cut post-encryption errors by 15%, by writing SQL-based data quality checks across 10,000+ banking records
- Restored unstable systems and reduced downtime, by tracing recurring failures through OBIEE server logs and Oracle databases
- Cut BI report response times by up to 30%, by optimizing the RPD repository and rewriting slow SQL queries
- Shipped Oracle SQL and XML changes to production with zero post-release defects, by owning change and release management in ServiceNow
- Lowered escalations by 25%, by resolving 20+ recurring incidents through structured root cause analysis

PROJECTS

Healthcare Claims Cost Intelligence Platform

github.com/ManojMareedu/Health-Care-Project | Live: healthcare-claims-cost-intelligence.streamlit.app

- Rebuilt an academic R/Jupyter healthcare claims analysis into a production Python system (ZenML, MLflow, DVC, pandera, FastAPI, Streamlit, Docker), by finding and fixing 5 real methodology bugs, including train/test leakage at the beneficiary level and a currency-parsing error that had silently dropped an entire state from the data
- Reached 78.9% accuracy, 0.753 macro-F1, and 0.954 ROC-AUC across 5 cost tiers with the best classifier (XGBoost), on a test set with zero beneficiary overlap with training
- Lifted charge-prediction R^2 from 0.26 to 0.87 and roughly halved the typical dollar error, by fitting gradient-boosted and random forest regressors instead of stopping at a linear baseline
- Made every prediction explainable, by adding SHAP feature attributions to the API and dashboard, plus Evidently AI drift monitoring and 51 automated tests

Predictive Analytics, Retail Demand Forecasting

github.com/ManojMareedu/Predictive-analytics-Data-Science-Project | Live: predictive-analytics-retail-sales.streamlit.app

- Rebuilt a graduate forecasting project (1M+ rows of IRI point-of-sale data) into a production system, by discovering the original ~81% accuracy came from features that were just arithmetic components of the target
- Reported an honest holdout R^2 of 0.635 instead of the inflated number, by moving evaluation to a temporal split (train on 2018–2021, test on unseen 2022 data) and explaining the gap rather than hiding it
- Wired up the full MLOps path around the model, including DVC-versioned data, MLflow tracking, FastAPI serving, and a Kubernetes deployment manifest, with CI on every push
- Delivered a public Streamlit dashboard with regional and brand-level breakdowns, live prediction, and a model comparison view

Credit Card Fraud Detection, Production ML System

github.com/ManojMareedu/Credit-Card-Fraud-Detection-Project | Live: credit-fraud-ml-dashboard.streamlit.app

- Built a full pipeline from data to deployment, using ZenML for orchestration, MLflow for experiment tracking, and DVC for data versioning
- Reached AUC 1.00, precision 1.00, and F1 0.9999 with the best model, on a synthetic dataset, so the near-perfect scores reflect that rather than real-world fraud rates
- Cut manual deployment to zero steps, by containerizing the inference service with FastAPI and Docker and wiring lint, build, and smoke tests into GitHub Actions
- Gave users a live view into their own risk, with a public Streamlit dashboard showing fraud probability, feature importance, and a plain-language risk interpretation

PUBLICATIONS

3D Deep Convolutional Neural Network for Detection of Anomalous Rigid and Affine Registrations in Big Data Brain MRI | IEEE

- Designed and deployed a 3D CNN (TensorFlow) for autonomous medical image assessment, reaching 99% accuracy and F1-score with GPU-optimized training on Google Colab
- <https://ieeexplore.ieee.org/document/9673318>

SKILLS

Languages & Core: Python, SQL, R, Bash

ML/DL: Scikit-learn, XGBoost, Gradient Boosting, Random Forest, HistGradientBoosting, PyTorch, TensorFlow, 3D CNNs, Computer Vision, Time Series Forecasting, Causal Inference, A/B Testing, Hypothesis Testing, statsmodels, Anomaly Detection

GenAI/LLM: LangChain, LangGraph, Agentic AI, Multi-Agent Systems, RAG, Prompt Engineering, Model Fine-tuning, Hugging Face Transformers, Sentence Transformers, LLMs, RAGAS, Responsible AI, Model Governance, Langfuse

Vector/Search: ChromaDB, FAISS, Pinecone, Weaviate, Vector Search

MLOps/Infra: MLflow, ZenML, DVC, Experiment Tracking, Model Registry, Feature Store, Evidently AI, pandera, Docker, Kubernetes, FastAPI, GitHub Actions (CI/CD), Git

Desktop & Packaging: pywebview, PyInstaller, watchdog, Inno Setup

Cloud: AWS, EC2, Lambda, S3, SageMaker, Glue, EMR, Google Cloud, BigQuery, Vertex AI

Data/BI: PySpark, Spark, Hadoop, Data Pipeline Engineering, Data Warehousing, Informatica ETL, Power BI, Tableau, Looker, Matplotlib, Seaborn, Plotly, MySQL, Oracle SQL, NoSQL

CERTIFICATIONS

AWS Certified Solutions Architect – Associate

Google Data Analytics

Google Business Intelligence

Google Deploy & Manage Generative AI Models (Trained)

Google Machine Learning Engineer (Trained)

EDUCATION

Master of Science in Business Analytics — Data Science

The University of Texas at Dallas, Jindal School of Management • GPA: 3.58 USA • 08/2022 – 05/2024

Bachelor of Technology in Electronics and Communication Engineering

Velagapudi Ramakrishna Siddhartha Engineering College • GPA: 3.23 India • 07/2017 – 07/2021

AWARDS & RECOGNITIONS

Scholar with Recognition

The University of Texas at Dallas

VOLUNTEERING & LEADERSHIP

The University of Texas at Dallas

01/2023 – 12/2025

President – Treasure Ministries

Naveen Jindal School of Management

12/2022 – 05/2023

Program Mentor